

Probability – 3

Random walks, Markov chains and limit theorems

Homework

11 сентября 2026 года

Theoretical minimum. Markov chains, transition matrices and stationary distributions; gambler's ruin; recurrence on $\mathbb{Z}$ and $\mathbb{Z}^2$ and transience on $\mathbb{Z}^3$; convergence of finite aperiodic chains and the coupling proof; branching processes and the extinction equation $f(s) = s$; the law of large numbers, the two Borel–Cantelli lemmas and concentration in high dimension.

1. (ISTCiM 2023) A beetle walks along the edges of a cube; on reaching a vertex it continues along one of the other two edges, chosen at random. Count the paths along which it returns to its starting vertex for the first time after exactly $2n$ steps.
2. (Putnam 2022) For $0 \leq p \leq \frac{1}{2}$ let $X_1, X_2, \dots$ be independent with $\Pr(X_i = 1) = \Pr(X_i = -1) = p$ and $\Pr(X_i = 0) = 1 - 2p$. For integers $b, a_1, \dots, a_n$ let $P(b, a_1, \dots, a_n)$ be the probability that $a_1 X_1 + \dots + a_n X_n = b$. For which p is $P(0, a_1, \dots, a_n) \geq P(b, a_1, \dots, a_n)$ always true?
3. (Мехмат, 2015) Does there exist a random variable ξ with integer values such that $\Pr(n \mid \xi) = \frac{1}{n}$ for every natural n ?
4. (Мехмат, 2025) Random variables ξ and η on one probability space have the same distribution, independence is not assumed, and $\xi - \eta$ has finite expectation. Must that expectation be zero?
5. (Высшая лига 2014) Prove that for all $\varepsilon, \delta > 0$ there is N such that for $n > N$ the probability that the scalar product of two independent uniform random unit vectors of $\mathbb{R}^n$ exceeds ε is less than δ .

6. (VJIMC 2019) Let p be an even non-negative continuous density and let $\xi_1, \dots, \xi_n$ be independent with density p . Put $X_0 = 0$ and $X_k = X_{k-1} + \xi_k$. Prove that the probability that $X_1, \dots, X_{n-1}$ all lie between X_0 and X_n equals $\frac{1}{n}$.
7. (S.T. Yau CSMC 2016) Let $N \geq 2$ and let X take values in $\{0, 1, 2, \dots\}$ with $\Pr(X \equiv k \pmod{N}) = \frac{1}{N}$ for all k . Compute $E(e^{2\pi i m X/N})$ for every integer $m \geq 1$, and use the answer to analyse the game in which N players each show a uniform number of fingers from 0 to 5 and the player numbered $S \bmod N$ is chosen.
8. Prove both Borel–Cantelli lemmas and deduce that a sequence of independent events with $\sum \Pr(A_n) = \infty$ happens infinitely often almost surely, while the converse fails without independence.
9. Prove the law of large numbers in the L^2 form, and prove the central limit theorem for ± 1 variables directly, by Stirling’s formula applied to the binomial coefficients.
10. Prove that an irreducible aperiodic finite chain converges to its stationary distribution by constructing a coupling, and give an example of a periodic chain for which convergence fails while the time averages still converge.